DATA ANALYSIS PROJECT

Restaurant Analysis & Performance

Exploratory Data Analysis (Python) + Business Questions (SQL) on restaurant data from around the world

9,551	141	15	145
restaurants	cities	countries	unique cuisines

Python · Pandas · Seaborn · MySQL · Jupyter

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SWIPE >

PROJECT OVERVIEW

What I set out to do

OBJECTIVE

Understand restaurant performance: which cuisines, cities and price tiers dominate, what drives ratings, and where the data can mislead.

DATASET

9,551 restaurants, 21 columns: location, cuisines, price range, rating, votes, delivery and booking flags.

TOOLS

Python (Pandas, NumPy, Matplotlib, Seaborn) for cleaning and EDA. MySQL for business questions on the cleaned table.

WORKFLOW

01 Clean — 02 Explore — 03 Query — 04 Insights

STEP 1 | DATA CLEANING

Preparing the data before any analysis

1

Fixed encoding and column names

Removed broken characters and converted every column to lowercase snake_case so Python and SQL use the same names.

2

Handled missing values

Only the cuisines column had nulls. They were filled with the most frequent value (mode).

3

Checked duplicates

0 duplicate rows and 0 duplicate restaurant IDs in the final data.

4

Treated outliers (IQR method)

Capped average_cost_for_two and votes at Q3 + 1.5 x IQR. This is why votes top out at 320.

5

Set correct data types

Converted flags and rating labels to categories, then exported cleaned_data.csv into MySQL for the SQL tasks.

THE DATASET AT A GLANCE

Key numbers behind the analysis

3.44
average rating
(rated restaurants)

22.5%
restaurants
not rated yet

25.7%
offer online
delivery

12.1%
offer table
booking

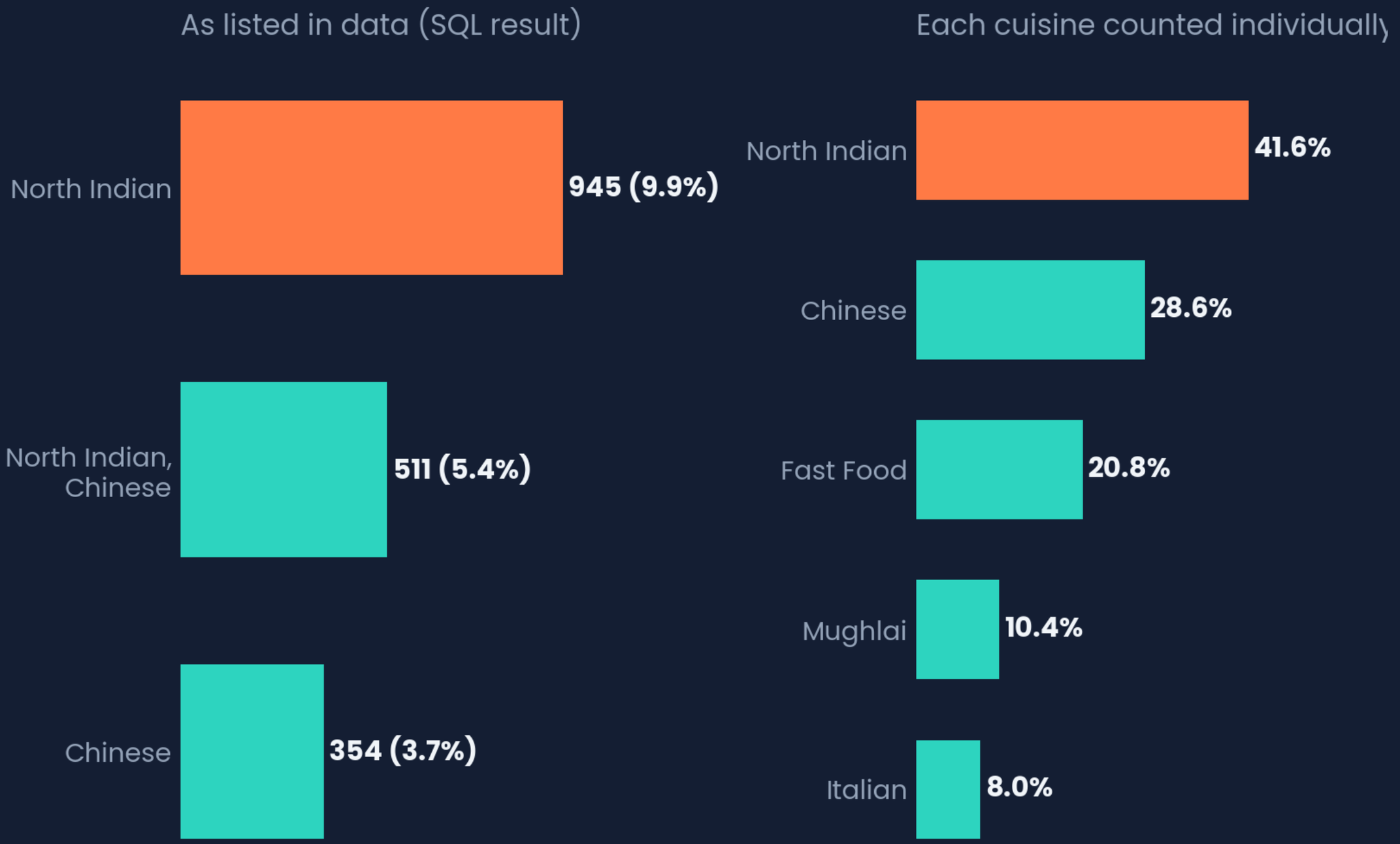
86
average votes
(median 31)

90.6%
of listings are
in India

Restaurants with **table booking** average **3.59** vs 3.41 without. Those with **online delivery** average **3.38** vs 3.47. (Rated restaurants only.)

SQL | TOP CUISINES

North Indian leads the menu

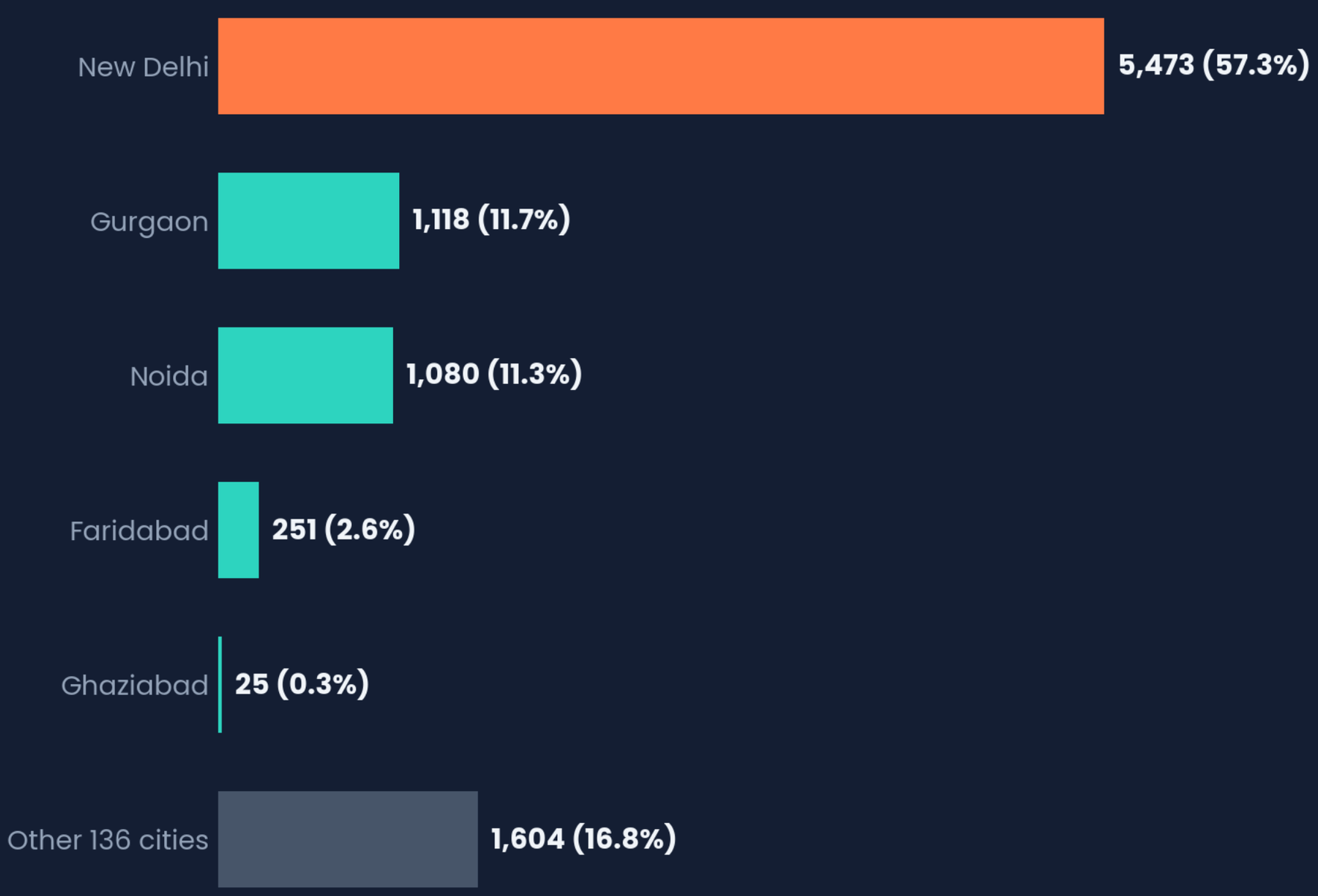


KEY INSIGHTS

- North Indian is #1 in both views: 9.9% as the exact listing, 41.6% once multi-cuisine listings are split.
- SQL treats "North Indian, Chinese" and "Chinese, North Indian" as different. Sorting the names merges them into 616 restaurants (6.4%).
- Individual shares add up to more than 100% because one restaurant can serve several cuisines.

SQL | CITY ANALYSIS

New Delhi has the most restaurants

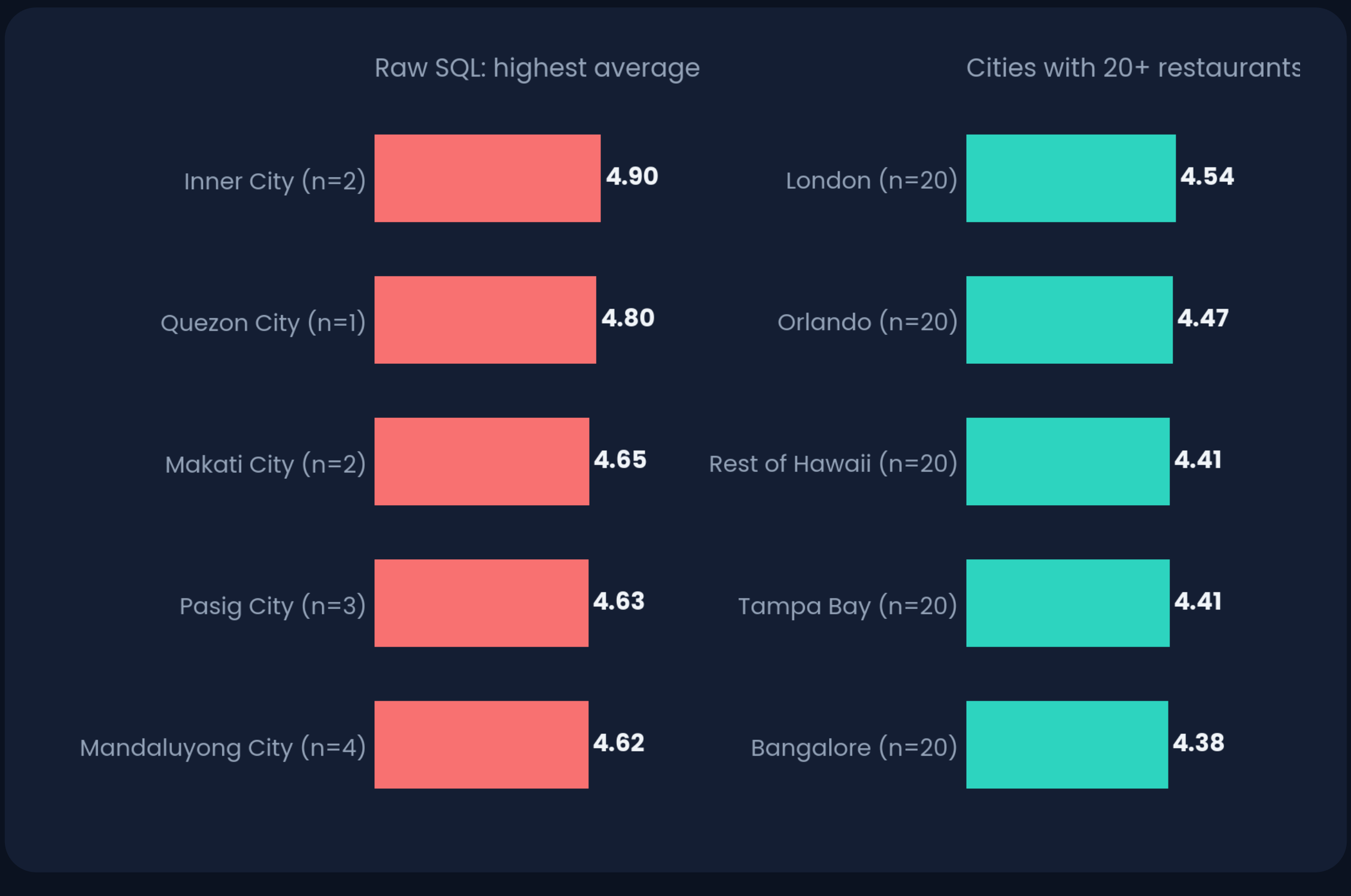


KEY INSIGHTS

- New Delhi alone has 5,473 restaurants, 57.3% of the entire dataset.
- The Delhi NCR cities (New Delhi, Gurgaon, Noida, Faridabad, Ghaziabad) make up 83.2% of all listings.
- The dataset is heavily India-centric, so city comparisons elsewhere rest on small samples.

SQL | CITY RATINGS

Highest rating? Check the sample size



KEY LESSON

- My SQL result: Inner City averages 4.9, but that comes from just 2 restaurants.
- Add HAVING COUNT(*) >= 20 and the picture changes: London leads with 4.54, followed by Orlando (4.47).
- Lesson: an average is only meaningful when the group is big enough to trust.

PYTHON | PRICE RANGE

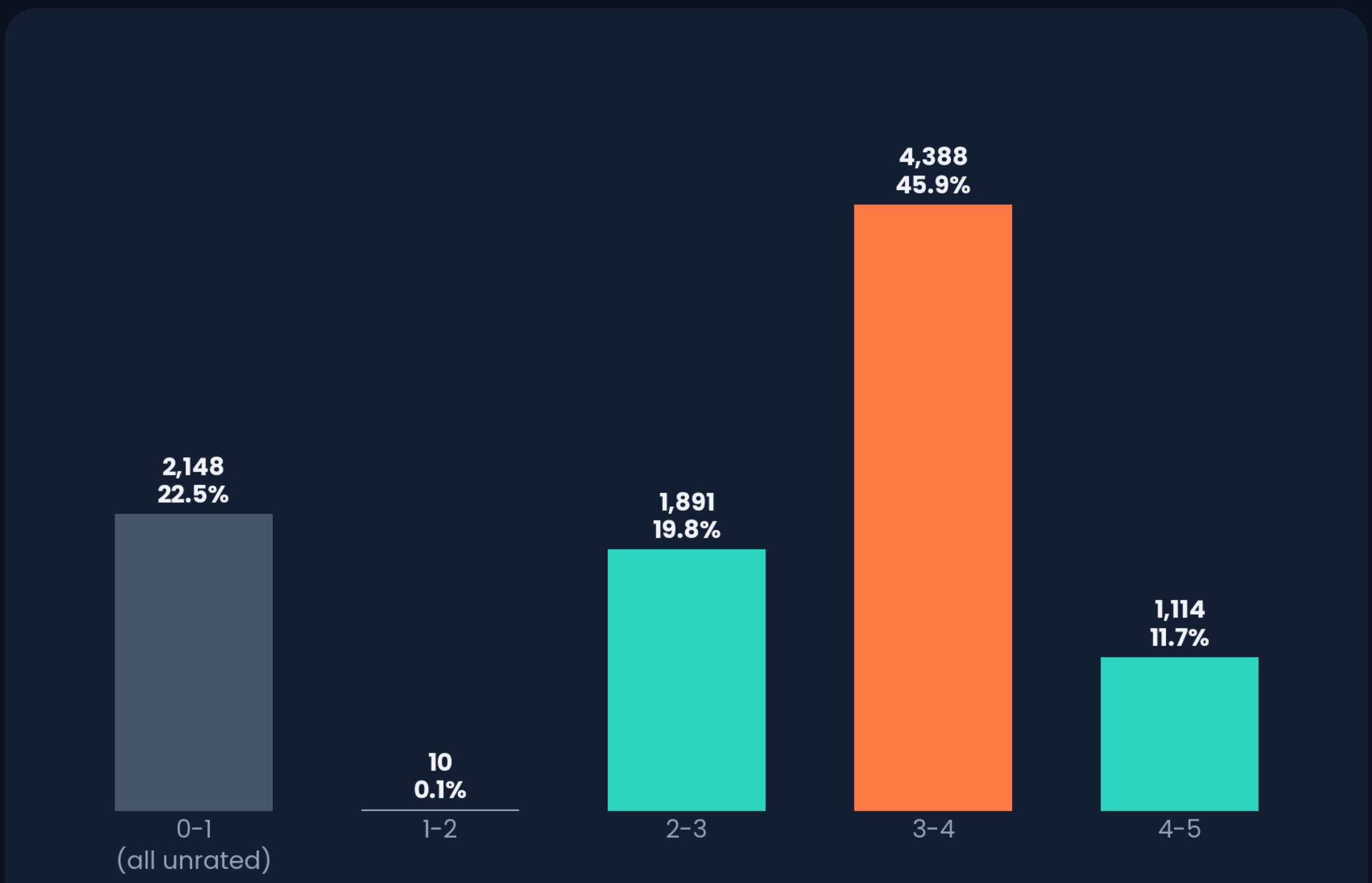
Most restaurants are budget-friendly



KEY INSIGHTS

- Price range 1 is the largest group: 4,444 restaurants (46.5%). Ranges 1 and 2 together cover 79.1%.
- Only 6.1% of restaurants are in the premium range 4.
- Higher price tiers earn better ratings on average (3.24 for range 1 vs 3.89 for range 4), though price does not cause quality.

Most restaurants score between 3 and 4

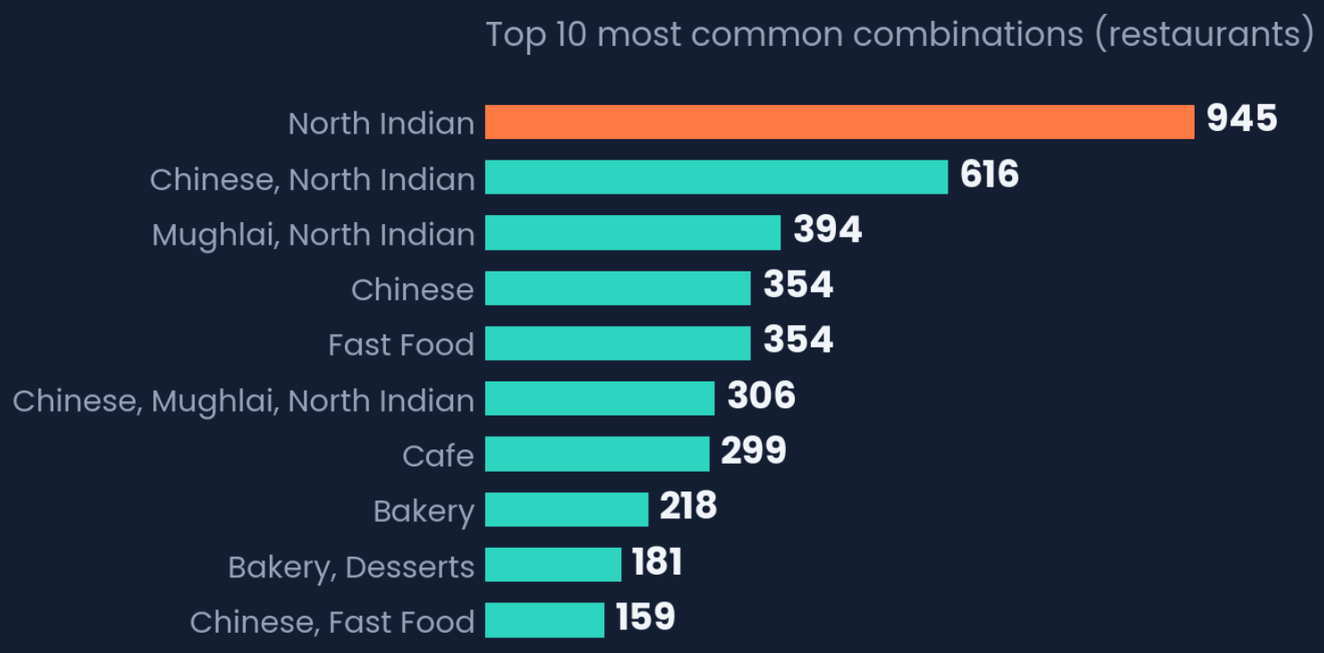


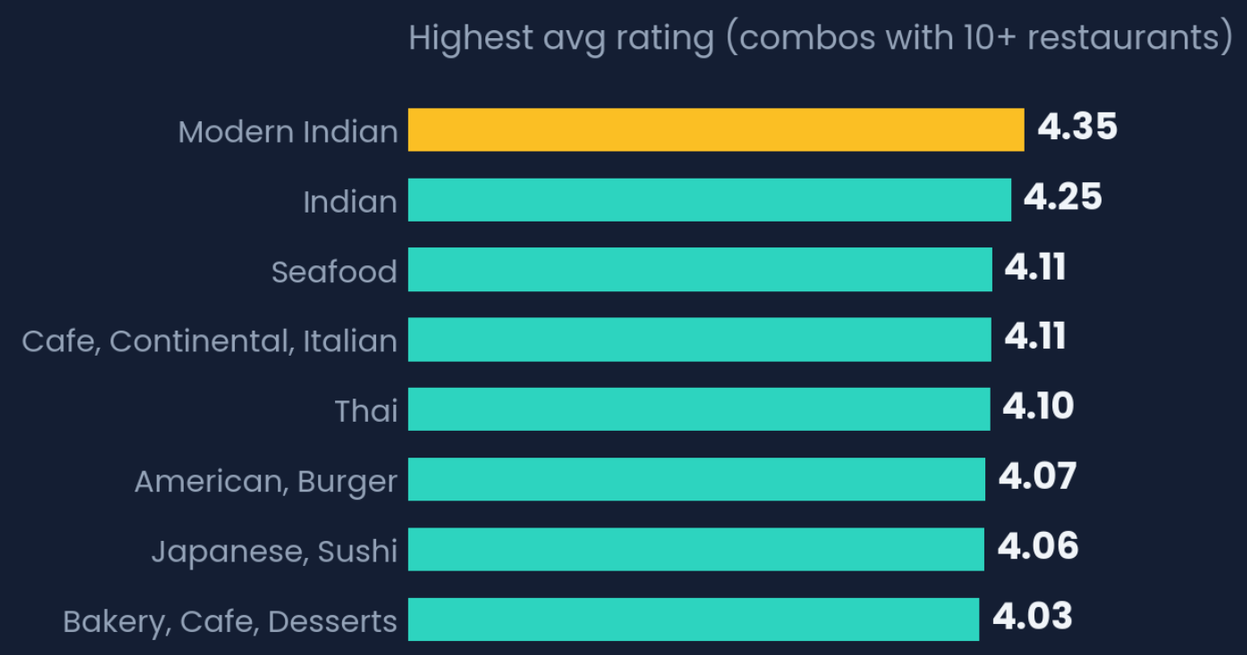
KEY INSIGHTS

- The most common range is 3-4 with 4,388 restaurants (45.9%).
- The 0-1 bar is not bad restaurants: all 2,148 are unrated (rating 0), which drags the overall mean down to 2.67.
- Average votes per restaurant: 86.06. The median is only 31, so most places get very few votes.

PYTHON | CUISINE COMBINATIONS

Which cuisine pairings win?



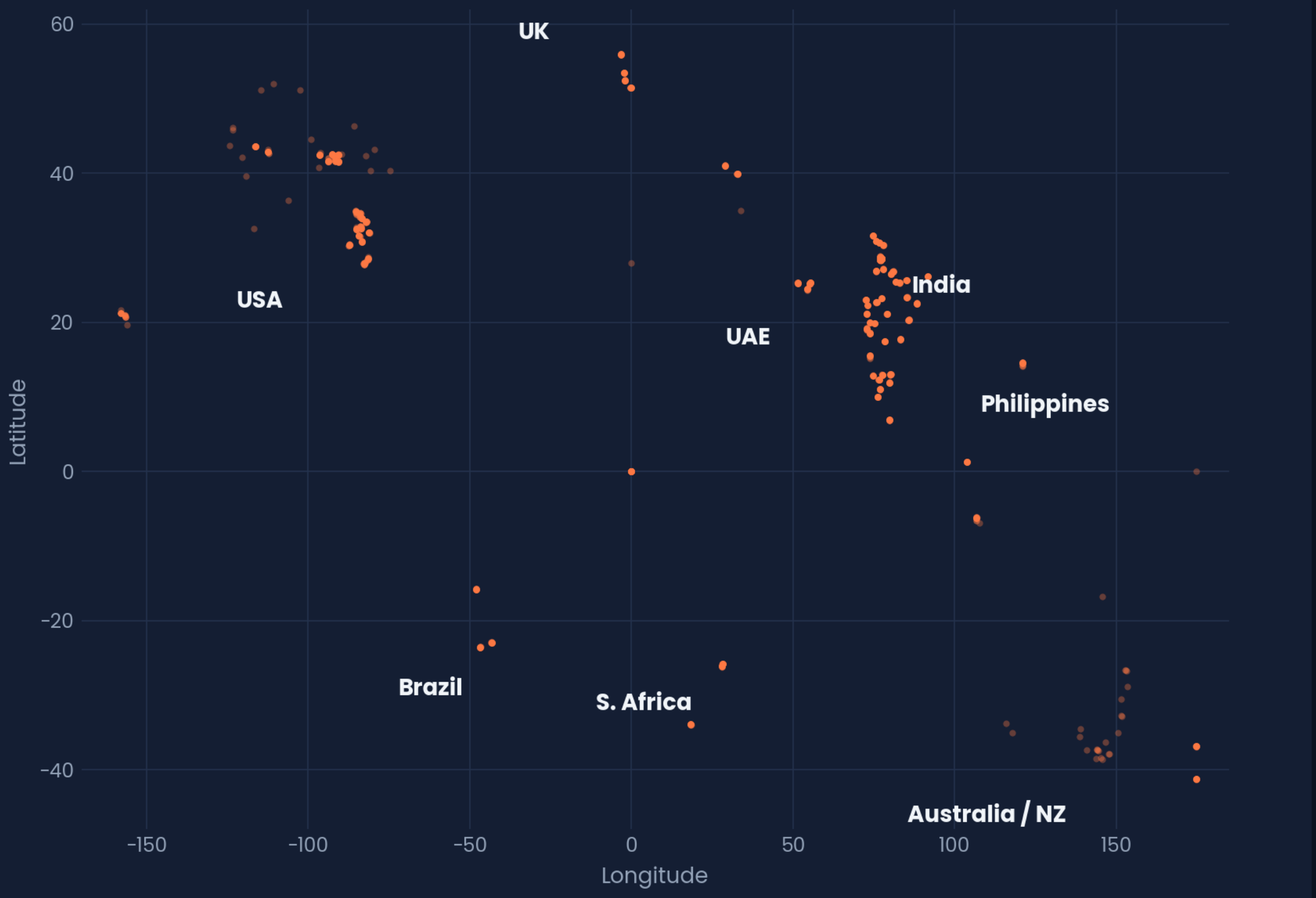


KEY INSIGHTS

- North Indian dominates, alone and paired with Chinese or Mughlai.
- Niche combos such as Modern Indian (4.35) and Seafood (4.11) rate higher, but each has under 20 restaurants.
- Popularity and quality are different things: the most common combos are not the best rated.

PYTHON | GEOGRAPHIC ANALYSIS

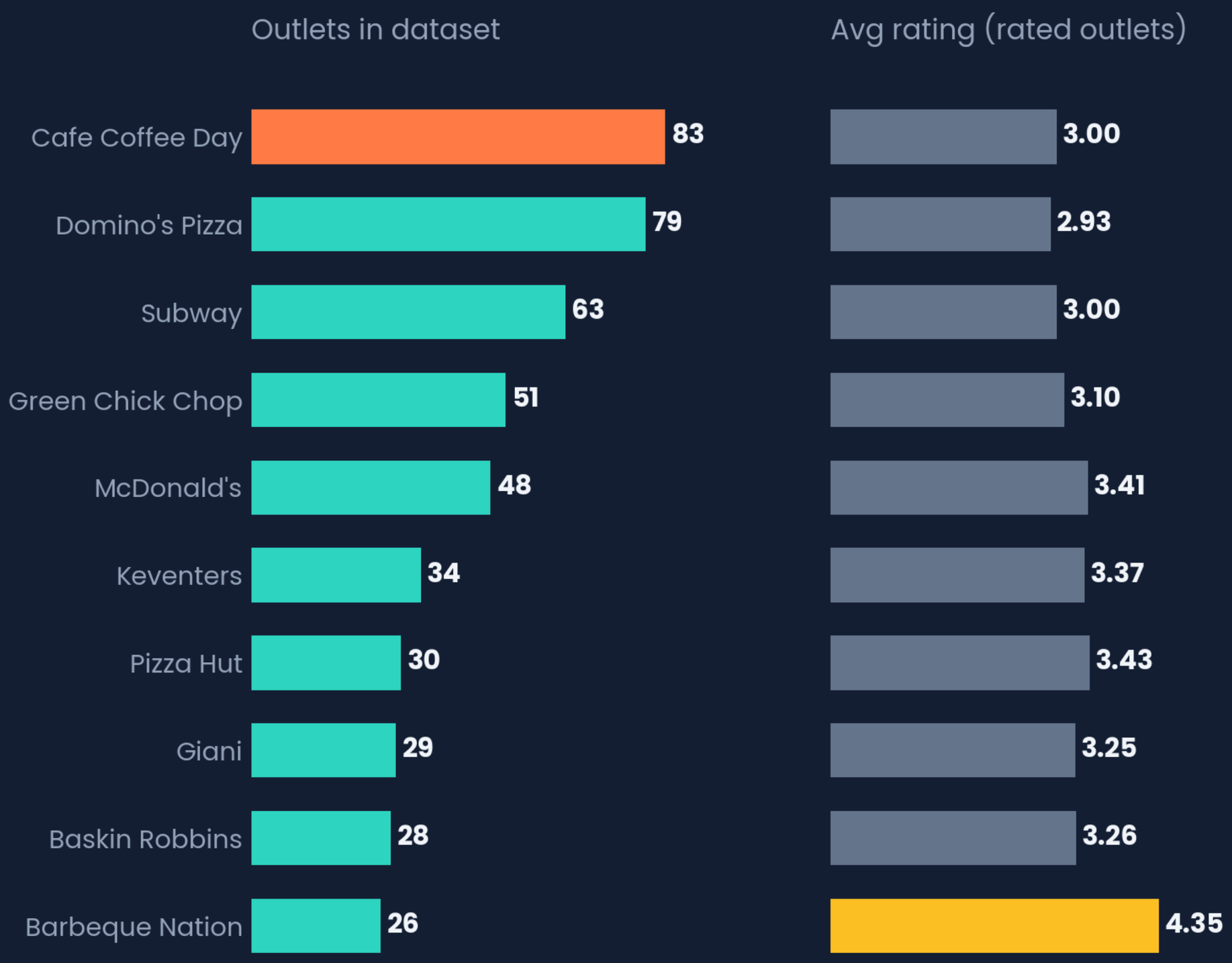
Where restaurants cluster



KEY INSIGHTS

- Dense clusters in India (90.6% of listings), the USA, the UK, Australia/NZ, Brazil and the Philippines.
- Inside India, the strongest cluster is the Delhi NCR region, matching the SQL city result.
- Large empty areas show where the dataset has no coverage, not where restaurants do not exist.

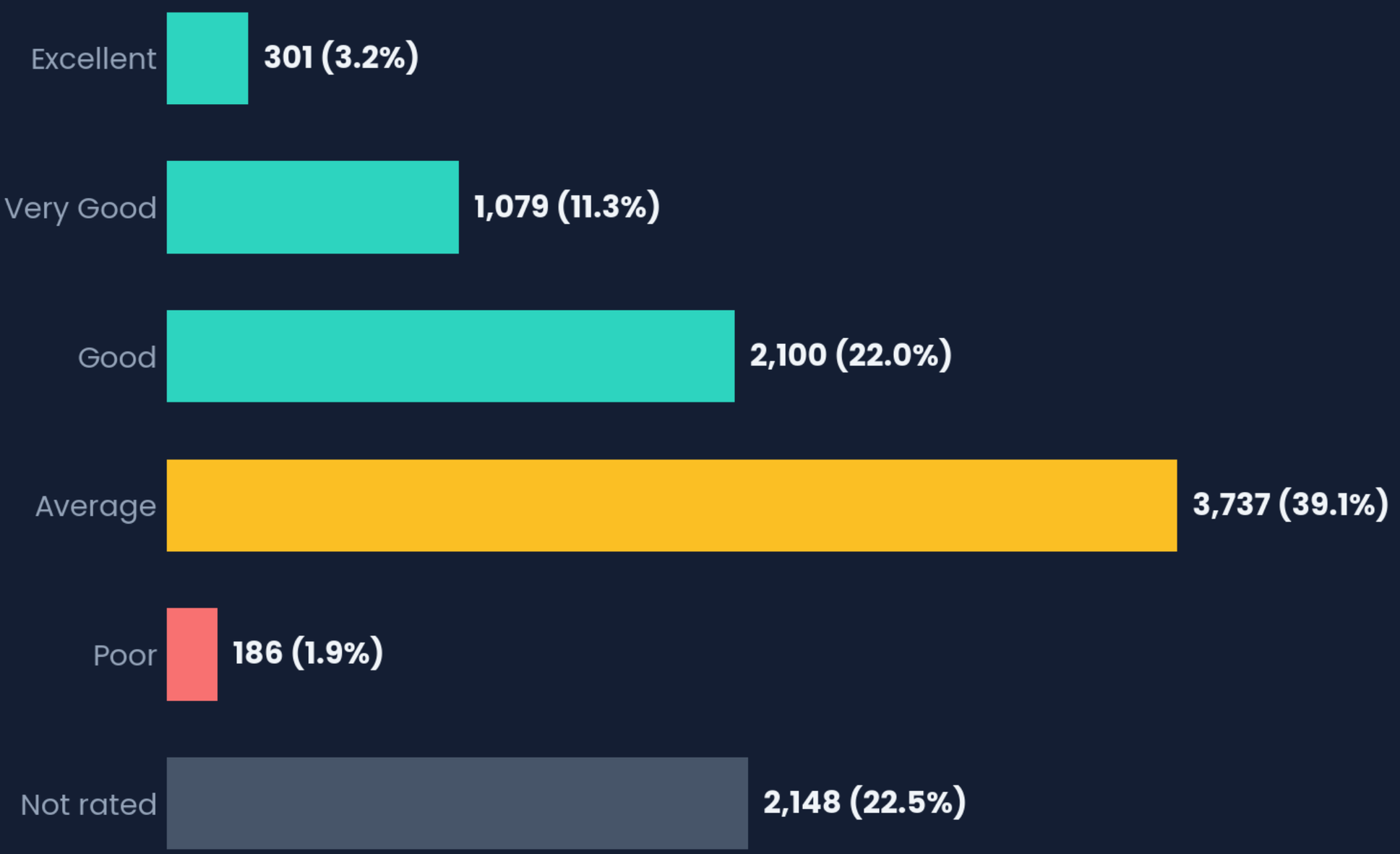
Big chains, very different ratings



KEY INSIGHTS

- 734 names appear more than once, covering 2,839 outlets. Cafe Coffee Day (83) and Domino's (79) lead.
- Barbeque Nation stands out: 26 outlets, a 4.35 average rating and about 261 votes per outlet.
- Scale does not guarantee quality: several of the biggest chains rate near or below 3.

What the rating labels say



KEY INSIGHTS

- Positive labels (Good, Very Good, Excellent) cover 36.4% of restaurants. Poor is only 1.9%.
- 'Average' is the single most common word (3,737), and 22.5% are 'Not rated'.
- Votes range from 0 to 320. The 320 is the IQR cap, shared by 1,128 restaurants, so it is not a true maximum.

BUSINESS TAKEAWAYS

What the data suggests

1

Delhi NCR is crowded

83% of listings sit in five NCR cities. New entrants need a clear niche or should look at under-served cities.

2

Digital features go with higher ratings

Restaurants with online delivery or table booking average higher ratings and get more votes. A signal worth testing, not a proven cause.

3

Budget is the volume market

79% of restaurants are in price ranges 1-2, while premium places rate higher but are rare.

4

Collect more reviews

22% of restaurants have no rating. Prompting diners to review would make the ratings far more reliable.

LIMITATIONS AND NEXT STEPS

What I would improve next

LIMITATIONS

- Votes were capped at 320, so popularity differences at the top are hidden.
- 22.5% of restaurants are unrated (0) and pull averages down.
- The 'review' analysis uses rating labels, not real review text, so true sentiment is untested.
- Chains are found by matching names, so variants like Giani and Giani's count separately.
- The dataset is 90%+ India, so global conclusions are limited.

NEXT STEPS

- Build a Power BI / Tableau dashboard
- Mine real review text with NLP
- Predict ratings with a regression model

Thank you!

Full Python notebook, SQL queries and cleaned dataset are on my GitHub. Link in the post.

What would you analyse next?

Feedback and suggestions welcome

Suraj Kapare

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